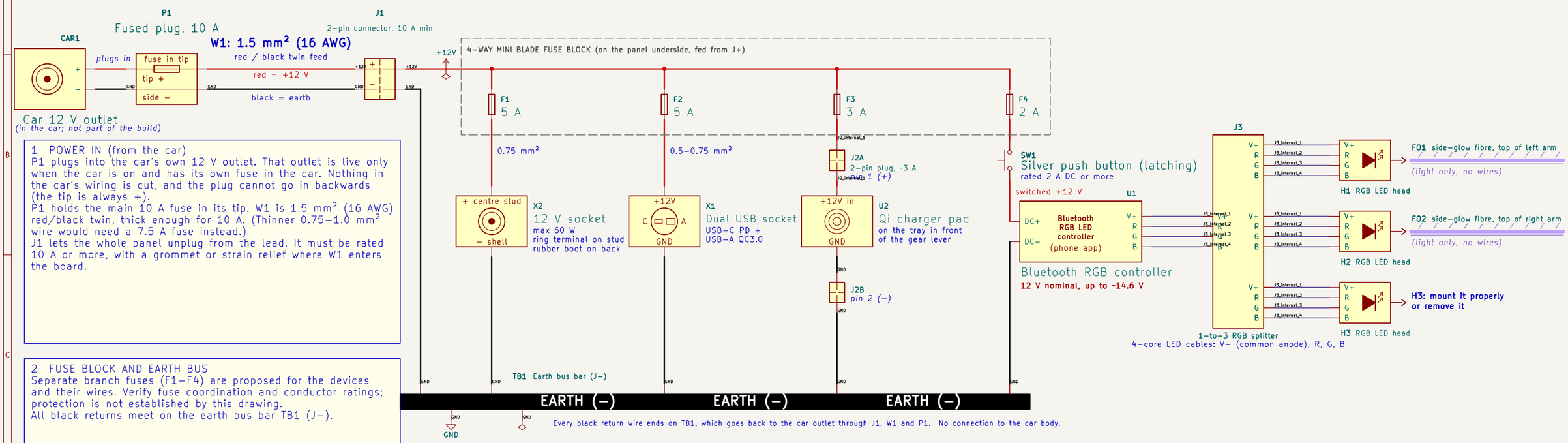
CUSTOM AFTERMARKET CAR CENTER CONSOLE UPGRADE

IMPROVED REDESIGN (not what was originally built)

Proposed changes to the console circuit: separate fused branches and clearer connections. Proposed circuit revision. This arrangement has not been installed.



1 POWER IN (from the car)
P1 plugs into the car's own 12 V outlet. That outlet is live only when the car is on and has its own fuse in the car. Nothing in the car's wiring is cut, and the plug cannot go in backwards (the tip is always +).
P1 holds the main 10 A fuse in its tip. W1 is 1.5 mm² (16 AWG) red/black twin, thick enough for 10 A. (Thinner 0.75–1.0 mm² wire would need a 7.5 A fuse instead.)
J1 lets the whole panel unplug from the lead. It must be rated 10 A or more, with a grommet or strain relief where W1 enters the board.

2 FUSE BLOCK AND EARTH BUS
Separate branch fuses (F1–F4) are proposed for the devices and their wires. Verify fuse coordination and conductor ratings; protection is not established by this drawing.
All black returns meet on the earth bus bar TB1 (J–).

LEGEND

- +12 V wire (red)
- earth / 0 V return (black)
- LED cable, 4-core (V+, R, G, B)
- light into a fibre (optical, no wires)
- one fuse block
- J2A / J2B = the two contacts of plug J2
- side-glow fibre

3 12 V SOCKET (X2)
Fuse F1 5 A, 0.75 mm² wire. Centre stud = +, shell = –. Ring terminal on the stud (not a push-on spade) and a rubber boot over the back. Label it "max 60 W".

4 USB SOCKET (X1)
Fuse F2 5 A, 0.5–0.75 mm² wire. One round socket with USB-C PD and USB-A QC3.0 (about 30 W + 18 W, typical figures). Its blue ring lights when it has power.

5 WIRELESS CHARGER (U2)
Fuse F3 3 A. The wireless charging pad sits on the tray in front of the gear lever and unplugs at J2 (2-pin plug, about 3 A). Its Qi module must accept 12–14.6 V directly.

6 RGB LIGHTS (the only switched circuit)
Fuse F4 2 A. The silver push button SW1 is the only control on the console: press it and the RGB lights come on. It switches the +12 V feed to the Bluetooth controller U1, so use a latching button rated 2 A DC or more. Colours and modes are set in the phone app.
U1 drives three aluminium RGB LED heads through the 1-to-3 splitter J3 (4 wires each: V+ common anode, R, G, B). H1 and H2 shine into the side-glow fibres along the tops of the two side arms. Mount H3 properly or remove it.
The USB socket, the 12 V socket and the Qi pad do not go through the button: they are live whenever the car's outlet is live.

PARTS (recommended ratings for the redesign)

P1	Fused 12 V cigarette plug, 10 A fuse in the tip
W1	Feed lead, 1.5 mm² (16 AWG) red/black twin
J1	2-pin car connector, 10 A or more, with grommet
F1–F4	4-way mini blade fuse block: 5 A, 5 A, 3 A, 2 A
X2	12 V accessory socket, max 60 W, rubber boot
X1	Dual USB socket: USB-C PD + USB-A QC3.0
J2	2-pin plug for the Qi pad, about 3 A
U2	Qi wireless charger pad, 12–14.6 V input
SW1	Silver push button, latching, 2 A DC or more
U1	Bluetooth RGB LED controller (phone app)
J3	1-to-3 splitter, 4-pin in-line plugs
H1–H3	Aluminium RGB LED heads, common anode
FO1, FO2	Side-glow optical fibre with fin (light only)
TB1	Earth bus bar or lever connector (J–)
CAR1	The car's own 12 V outlet (not part of the build)

PROPOSED WIRING PRACTICES (verify ratings before implementation)
- Red = +12 V and black = earth, everywhere.
- One wire per crimp barrel, with the barrel matched to the wire size. No doubled wires.
- Spade connectors only on flat tabs; a ring terminal on the 12 V socket's centre stud.
- A rubber boot over the back of the 12 V socket, so the bare stud can never touch the seat-belt buckle.
- Sleeving and clips on every run, and grommets wherever a cable passes through the board.
- Keep every cable away from the seat rails, the seat-belt buckle, the gear lever and the EPB / drive-mode / EV switches.
- Mount the third LED head (H3) properly or remove it.
- No car-body earth and no fuse-box tap, unless I confirm that is how it was originally powered.
- Optional: if the new button has a 12 V LED ring, wire the ring from the switched side to earth.

POWER BUDGET at 12 V: USB up to ~4.5 A, Qi pad ~1.5–2 A, lights ~0.5–1 A = about 7 A before anything is plugged into the 12 V socket. These are estimates, not measured currents. Verify the combined load against the outlet, wiring and connector ratings; the fuse is not a load regulator.

ABOUT THIS SHEET
I designed the panel and planned its circuit before fabrication. This present-day drawing proposes a revised circuit; it is not the installed wiring and has not been built or validated. For the installed arrangement, see console_as_built.pdf. Ratings are recommendations; wire sizes and values are approximate.
From the original build: the only control is a silver push button that turns on the RGB LEDs. The red-ringed part is a 12 V socket, not a switch.

STILL TO CHECK before building this version
- The car's outlet: that it is off when the car is off, and its rating (usually 120 W / 10 A on a ~15 A fuse; check the owner's manual).
- U1 and the Qi module must accept up to ~14.6 V (the original controller's label says 5–12 V).
- The original button latches; its current rating is unknown. Specify a switch rated 2 A DC or more.

Removable centre-console panel
Proposed revision; not installed or physically validated
Drawn 2026-10-03
IMPROVED REDESIGN, spec section 5.2 – NOT what was originally built
Console circuit: proposed revision
Sheet: /
File: console_improved.kicad_sch

Title: Custom Aftermarket Car Center Console Upgrade – improved

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